

Dreadnought Arizona, Near Completion, to Be Biggest Warship Afloat



What Would Happen if
Greatest of Floating Forts
Got Into a Battle with
The Queen Elizabeth.

U.S. BOAT INVINCIBLE WHEN IN ITS RANGE

By HENRY BATE.



BATTLESHIP 39—the Arizona—58 per cent. completed," was the official announcement made in Washington several days ago.

To the average American the figures mean little or nothing; but to the naval officers and men in the New York Navy Yard, where Uncle Sam's new big superdreadnought is being constructed, the announcement carries joy and satisfaction. It means that American naval officers are breaking records in battleship construction.

So consistently has work on the big fighting ship progressed since her keel was laid March 16, 1914, that her constructors are several months ahead of their schedule. They will be able to launch her early next June, long before she is due to take the water.

To naval officers in general the official announcement will carry a wave of proud anticipation—anticipation of the time when the Arizona, acknowledged the queen of fighting craft, will show her colors and take her place in the iron band being forged along America's seacoasts.

With the completion of the Arizona America will have a division of ultra-modern superdreadnoughts carrying heavy armor and big fourteen-inch guns, asserted by many experts to be the masters of any firing pieces afloat, the fifteen-inch cannon of the new British battleship Queen Elizabeth and her sister ships not excepted.

The Largest Ever Built.

The Arizona is the largest battleship built or under construction, here or abroad, unless Great Britain or Germany has secretly laid down sea monsters beyond present conception since the war started.

The Arizona will displace 31,500 tons light and more than 32,000 tons when fully equipped and manned. The nearest approach to such a tonnage is made by the four new dreadnoughts being turned out for Germany and the four under construction for Japan. These ships will displace about 30,000 tons, far short of the Arizona and her sister ship, the Pennsylvania, recently launched. From the time she first cuts the water the Arizona will reign supreme until the first of three new American battleships, now not even laid down, is launched. The California, the Idaho and the Mississippi each will displace more than 32,000 tons light.

American naval experts are confident the Arizona, despite the failings along certain lines when compared to the "brag boats" of other navies, will prove a match for any afloat.

June 19 has been set officially as the day when the Arizona first leaves her gigantic sides in the waters of Wallabout.

At present she stands a towering structure above the Navy Yard in her building ways. Like

a modern Gulliver, she slumbers while the thousand-and-one workers of the Navy Yard swarm over her and over the timbers that, thread-like, hold her in place.

Already most of her powerful turrets have been set in place. In bright vermilion of lead paint, they form a curious contrast to the battleship gray already spread over the ship to the main deck level.

Rear Admiral Usher in Charge.

As the head of the local Navy Yard, Rear Admiral Nathaniel R. Usher is in charge of her construction. Under him, however, Naval Constructor Robert Stocker and Naval Constructor John E. Bailey are responsible for the putting together of the vast hull of the ship, and Lieutenant Commander Louis C. Richardson and Lieutenant Commander Stafford H. R. Doyle of the powerful engines and their auxiliaries.

The huge engines, of the Parsons turbine type, are being constructed in the machine shops of the yard, of which Commander Richardson is in charge. Six massive turbines with their thousands of delicate steam blades are being rushed to completion. Never has a Government plant undertaken such a huge task.

There is good reason to welcome the Arizona. A glance at her size and fighting qualities immediately shows her to be incomparable. Although built along the lines of the Pennsylvania, she is a few feet larger and a few hundred tons heavier than even that ship.

On the waterline she will measure 600 feet and will have an over-all length of 608 feet. Her beam is ninety-seven and a half feet and her mean draft almost twenty-nine feet. She will have five decks, with a bridge deck and two platforms. She will have twenty-four main transverse watertight compartments and about 400 compartments in all. She will develop about 35,000 horsepower and will be able to make almost twenty-two knots.

Armored on New Plan.

The Arizona is armored according to a new and entirely American plan first inaugurated in the battleships Oklahoma and Nevada, now about to go into commission.

Instead of plates of armor of varying thickness from stem to stern, the new American superdreadnoughts are almost devoid of armor except along the waterline and around the big guns.

Beginning many feet back of the prow, the Arizona will have a thick belt of armor eighteen feet wide, and varying from eleven to sixteen inches thick. This belt will reach eight feet below the waterline and will stretch along the ship for about 400 feet. There will be little or no armor either forward or aft of this belt on each side.

Such distribution of armor marks the depart-

The Arizona.

Tonnage, 31,500.
Length, water line, 600 feet.
Breadth, water line, 97½ feet.
Speed, 21.5 knots.
Horsepower, 35,000.
Belt armor, 16 inches.
Gun armor, 18 inches (?).
Armament—Primary, 12 14-inch guns;
secondary, 22 5-inch guns; 4 21-inch tor-
pedo tubes.
Complement—Officers, 65; men, 937.
Fuel, 2,322 tons of oil.
Cost, \$13,000,000.
Her main guns will throw a broadside of
8½ tons of metal almost 20 miles, and can
be used effectively at 13 miles.

ure in battleship protection. It is an "all or none" idea which the General Board has conceived to be the correct solution of the needs of the Navy's chief fighters.

Heaviest Armor Known.

The armor on the Arizona will be heavier than that on any foreign battleship, but it will be concentrated exactly where it is needed and nowhere else. It is said the Arizona's turret armor—the sloping plates over the tops of the guns—will be no less than eighteen inches thick. The conning tower will be protected by armor almost as thick.

The deck armor will be laid on only three inches thick, but it will be of the hardest steel obtainable—vanadium steel—and will be lapped so as to make it impenetrable.

The interest in any battleship centres in her guns. The guns of the Arizona will can invite inspection. Twelve in number, they can bite as well as bark. The fourteen-inch naval gun of the United States will fire a 1,400-pound projectile 14,000 yards so effectively that no battleship afloat could stand up under the combined fire of the Arizona's major batteries, manned by the crack gunners the American Navy breeds.

Guns Also Concentrated.

The big guns of the Arizona are arranged in four turrets of three guns each. The Arizona and the Pennsylvania will be the first American ships so constructed. The General Board, as well as

concentrating their armor, are concentrating gun-fire aboard our new big ships. The Arizona can fire six guns ahead or astern or all twelve on either broadside. That broadside means 16,800 pounds of metal. With all her boasted navy, not one of Great Britain's ships can deliver that much destruction at once.

Besides her heavy guns, the Arizona will carry twenty-two five-inch rapid-fire guns. These form what is known as the torpedo defense.

The big battleship will carry sixty-five officers, a crew of 883 and a marine detachment of seventy-four men, a population equal to that of a small town.

The Arizona is an oil-burning ship. For that reason she is able to do away with the many funnels necessary to big coal burners. She has only one funnel, amidships. Her oil bunkers will carry 694,830 gallons of oil, or more than 2,322 tons. Think of what a great cruising radius she has!

Arizona vs. Queen Elizabeth.

It might be interesting to figure out what might happen if the Arizona and the Queen Elizabeth were matched.

In the first place, with her twenty-five knot speed the British ship would have the advantage of making or refusing fight. Assume that the two ships are speeding along on courses which eventually might intersect.

At about 36,000 yards, or about fifteen miles, the lookouts in the fighting platforms of the Arizona's basket masts would identify the fighting tops of the Queen Elizabeth as an enemy's warship. It would be many minutes later before the British seamen would know of the American warship.

The American basket mast, with its observation platform on top, is many feet higher than the observation platform of foreign battleships. To-day's sea fights are at much greater ranges than heretofore were thought possible. In the respect of getting the range and being ready to fire upon an enemy, the American ships are supreme.

Having sighted the Queen Elizabeth, the Arizona might have the chance to change her course and sneak away. Let's suppose she keeps on, looking for trouble and ready for it.

The Queen Elizabeth's chance of hitting the Arizona would be very small at 31,000 yards, or twelve miles. A moving battleship is a poor target

The U. S. S. Arizona,
Under Full Steam,
Firing a Salvo—
Drawn from the
Designer's Sketch.

twelve miles away. The fact that a ship's guns can shoot twelve or fourteen or sixteen or even twenty miles effectively is not of much importance when the target is a spot on the horizon and moving at twenty-two knots.

British Ship Opens Fire.

It is not unlikely that the Queen Elizabeth would open fire at 16,000 yards or more. There is little chance her fire would be answered. The American fourteen-inch gun, although it has a dangerous range of more than twelve miles, is not considered an effective weapon much beyond 14,000 yards. Our ships are built to do their fighting much closer—say at 10,000 and 12,000 yards.

At 16,000 yards the gunners aboard the Queen Elizabeth would have to be very lucky to land a shot on the Arizona. Should one of their shots tell, it would do little damage to either the side armor, the turrets or to the unprotected parts of the vessel. If it hit the deck it might pierce into the ship, putting several compartments out of commission. If it hit the belt armor it would have little effect. If it hit the fore and aft unprotected parts of the ship it might destroy one or more watertight bulkheads.

In other words, the shots the Queen Elizabeth could put against the Arizona while the American ship is outranged would not necessarily put the Arizona out of commission. It is just what she is built to take—punishment.

At 14,000 yards the American gunners would open fire. Their shots, crashing against the thinner armor of the British ship, might either render her hors de combat or force her to seek safety in her superior speed. We might have the picture of the Queen Elizabeth changing her course and disappearing over the horizon, raked by American gunfire as long as she is in range and replying with her eight monster fifteen-inch pieces.

Irresistible at Close Range.

Should the Queen Elizabeth close in with the Arizona, even the most skeptical critic of the American Navy concedes the fight would lie with the American ship. The 500 pounds difference between the shells of the British and American guns is vastly overbalanced by the difference in the thickness of armor. There also is the fact that for every eight 1,950-pound projectile fired by the British twelve 1,400-pound projectiles would be fired by the Arizona. The comparison stands: Total weight fired, British 18,000 pounds, American 16,800.

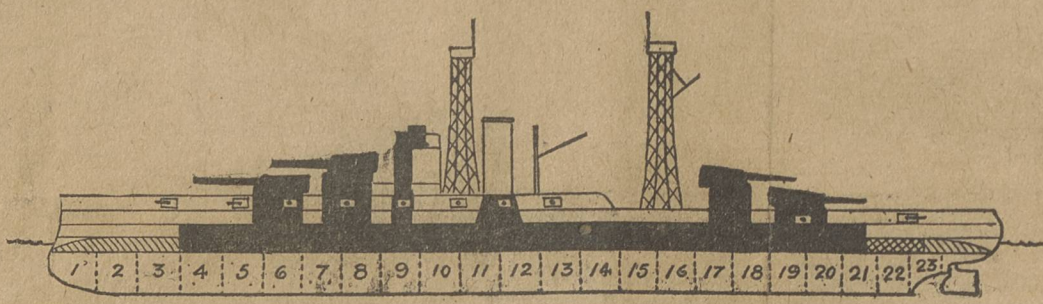


Diagram of the Arizona, Showing How the Armor (Black Portion) Is Massed.